

1 -

	$\lim_{u \rightarrow c} (f(u) + g(u))$	$\lim_{u \rightarrow c} (f(u) - g(u))$	$\lim_{u \rightarrow c} (f(u)g(u))$	$\lim_{u \rightarrow c} \frac{f(u)}{g(u)}$
$\lim_{u \rightarrow c} f(u) = a$ $\lim_{u \rightarrow c} g(u) = b$	$a + b$	$a - b$	ab	$\frac{a}{b}$

2 -

	$\lim_{u \rightarrow c} (f(u) + g(u))$	$\lim_{u \rightarrow c} (f(u) - g(u))$	$\lim_{u \rightarrow c} (f(u)g(u))$	$\lim_{u \rightarrow c} \frac{f(u)}{g(u)}$
$\lim_{u \rightarrow c} f(u) = 0$ $\lim_{u \rightarrow c} g(u) = b$	$0 + b = b$	$0 - b = -b$	$0 \times b = 0$	$\frac{0}{b} = 0$

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	$\lim_{u \rightarrow c} (f(u) + g(u))$	$\lim_{u \rightarrow c} (f(u) - g(u))$	$\lim_{u \rightarrow c} (f(u)g(u))$	$\lim_{u \rightarrow c} \frac{f(u)}{g(u)}$
$\lim_{u \rightarrow c} f(u) = \pm\infty$ $\lim_{u \rightarrow c} g(u) = b$	$\pm\infty + b = \pm\infty$	$\pm\infty - b = \pm\infty$	$+\infty, b > 0$ $-\infty, b < 0$	$+\infty, b > 0$ $-\infty, b < 0$

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	$\lim_{u \rightarrow c} (f(u) + g(u))$	$\lim_{u \rightarrow c} (f(u) - g(u))$	$\lim_{u \rightarrow c} (f(u)g(u))$	$\lim_{u \rightarrow c} \frac{f(u)}{g(u)}$
$\lim_{u \rightarrow c} f(u) = \pm\infty$ $\lim_{u \rightarrow c} g(u) = 0$	$\pm\infty + b = \pm\infty$	$\pm\infty - b = \pm\infty$	indeterminado	$\pm\infty, \text{ se } \lim_{u \rightarrow c} g(u) = 0^+$ $\pm\infty, \text{ se } \lim_{u \rightarrow c} g(u) = 0^-$

$\frac{f(u)}{g(u)} =$
 $f(u) \times \frac{1}{g(u)}$



$\lim_{u \rightarrow c} g(u) = 0^+$, então $\lim_{u \rightarrow c} \frac{1}{g(u)} = +\infty$

$\lim_{u \rightarrow c} g(u) = 0^-$, então $\lim_{u \rightarrow c} \frac{1}{g(u)} = -\infty$

Exercício

Dar exemplos de funções f e g tal que:

$$\lim_{x \rightarrow 0} f(x) = +\infty$$

$$\lim_{x \rightarrow 0} g(x) = 0$$

$$\lim_{x \rightarrow 0} f(x)g(x) = 0, 1, -1, +\infty, -\infty, \text{ não existe}$$

$\lim_{x \rightarrow 0} (f(x)g(x))$	$f(x)$	$g(x)$	$\lim_{x \rightarrow 0} f(x)$	$\lim_{x \rightarrow 0} g(x)$	$f(x)g(x)$
0	$1/x$	x^3	$+\infty$	0	x
1	$1/x$	x	$+\infty$	0	1
-1	$1/x$	$-x$	$+\infty$	0	-1
$+\infty$	$1/x^2$	x	$+\infty$	0	$1/x$
$-\infty$	$1/x^2$	$-x$	$+\infty$	0	$-1/x$
não existe	$1/x^2$	$x \sin(\frac{1}{x})$	$+\infty$	0	$\frac{\sin(1/x)}{x}$

